

# Robot Arm Kinematics

Prof. Gerardo Flores

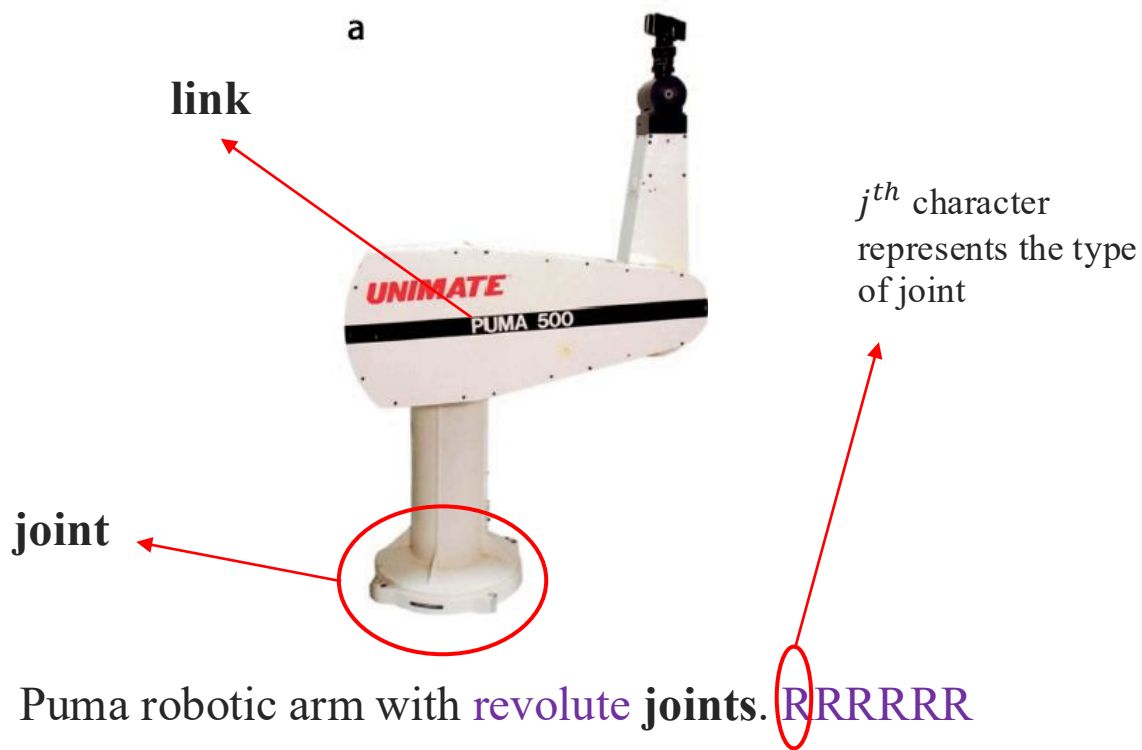
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Robotics and Automation course,  
TAMU



# Describing a robot arm

Kinematics is the branch of mechanics that studies **the motion of a body**, or a system of bodies, **without** consideration given to its **mass or the forces** acting on it.



[https://youtu.be/k86D\\_kEtFTU?si=aOnIauINe\\_b2YMcM](https://youtu.be/k86D_kEtFTU?si=aOnIauINe_b2YMcM)



The Stanford arm with one **prismatic joint**. **RRPRRR**

<https://youtu.be/O1oJzUS1TeY?si=G4c35PffXhkWqxGr>

# Describing a robot arm

What is the goal?

- The objective is to represent the position and orientation of any part of the robot, typically the gripper (or end effector), relative to a reference frame, usually the inertial or fixed frame.
- To achieve this, we must:
  1. Define multiple coordinate frames along various parts of the robot.
  2. Establish the homogeneous transformations between these frames.
  3. Calculate the homogeneous transformation that relates the inertial frame to the end effector's tool frame.

# Denavit-Hartenberg notation

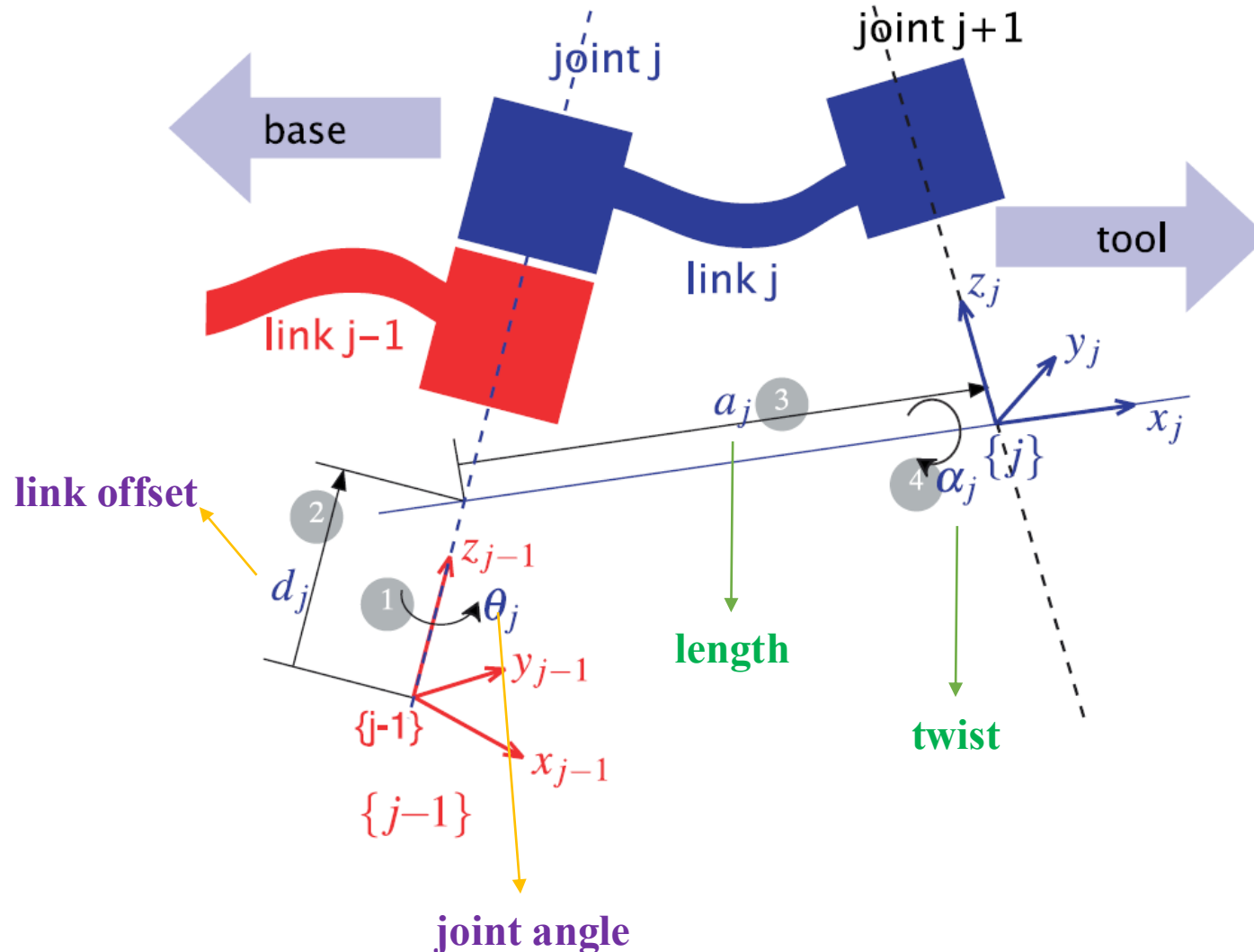
A systematic way of describing **the geometry** of a **serial chain of links and joints** was proposed by Denavit and Hartenberg in 1955 and is known today as *Denavit-Hartenberg notation*.

For a manipulator with  $N$  **joints** numbered from **1 to  $N$** , there are  $N+1$  links, numbered from 0 to  $N$ .

**Joints  $\rightarrow$  1 to  $N$**   
**Links  $\rightarrow$  0 to  $N$**

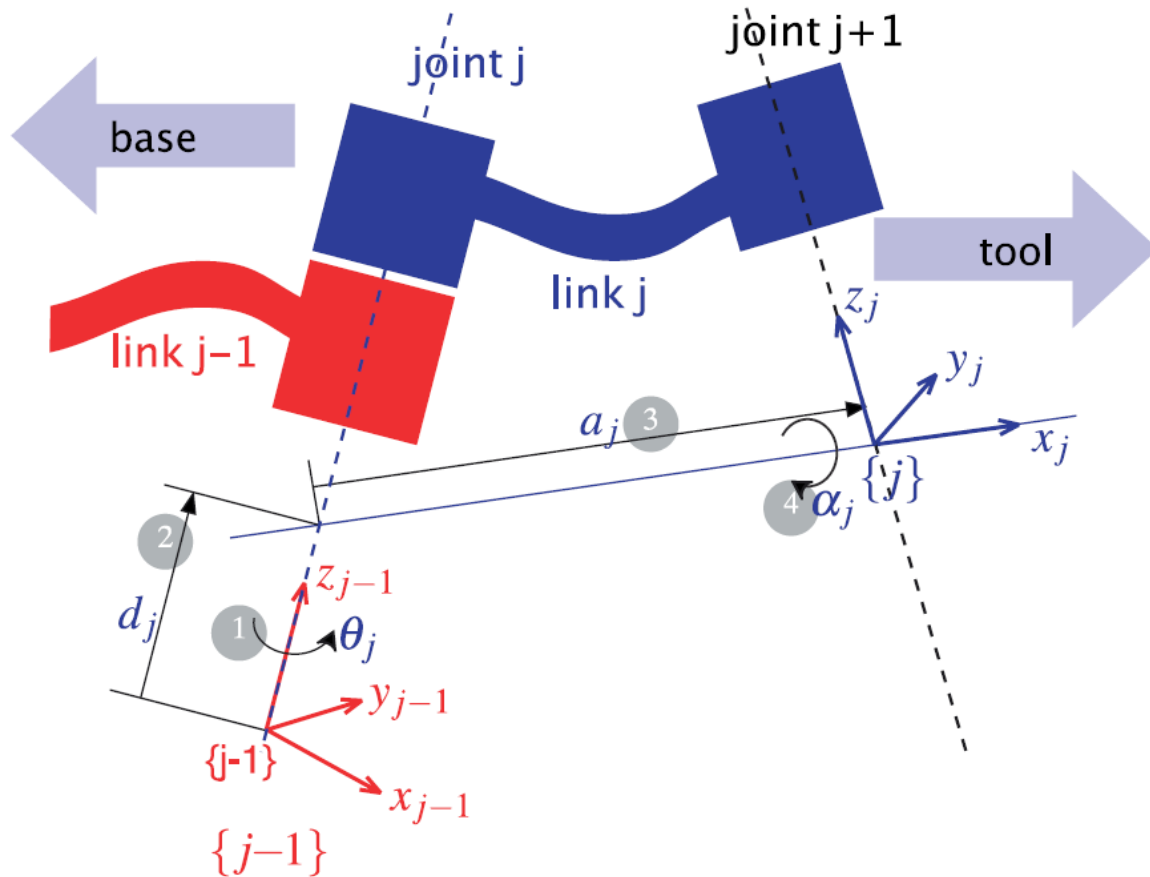
Link 0 is the base of the manipulator and link  $N$  carries the end effector.

# Definition of standard Denavit and Hartenberg link parameters

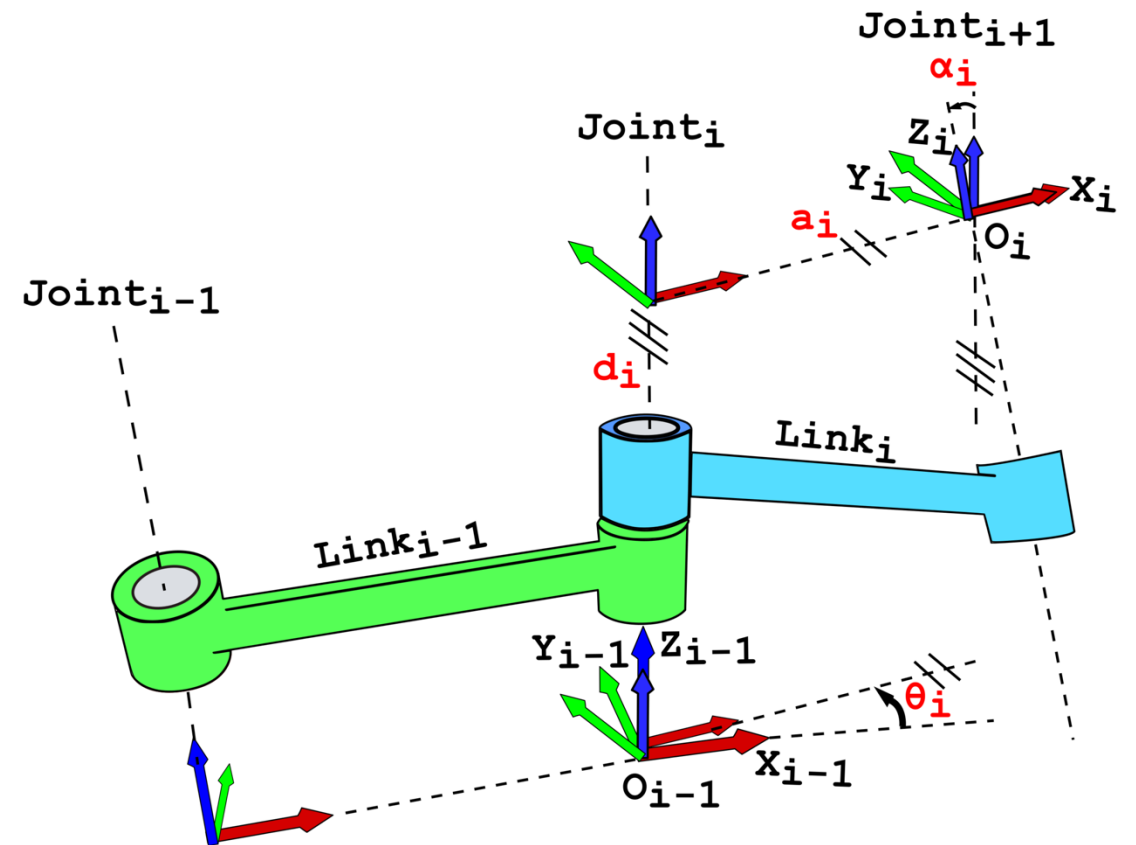


- The numbers in circles represent the **order** in which the elementary transforms are applied.
- Joint  $j$  connects link  $j-1$  to link  $j$  and therefore joint  $j$  moves link  $j$
- A **link** can be specified by two parameters, its **length**  $a_j$  and its **twist**  $\alpha_j$ .
- The **link offset**  $d_j$  is the distance from one link coordinate frame to the next along the axis of the joint.
- The **joint angle**  $\theta_j$  is the rotation of one link with respect to the next about the joint axis.

# Denavit and Hartenberg parameters

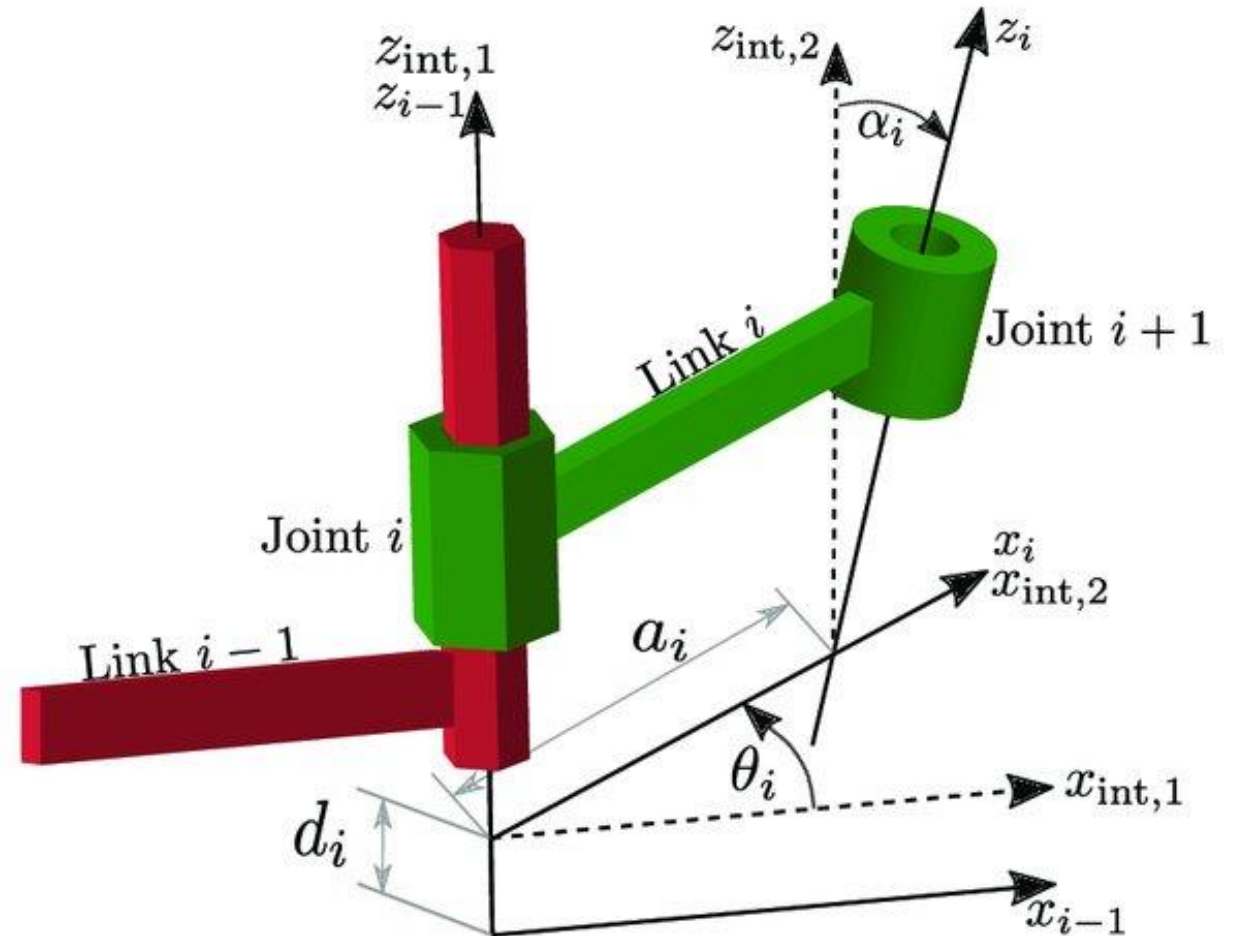
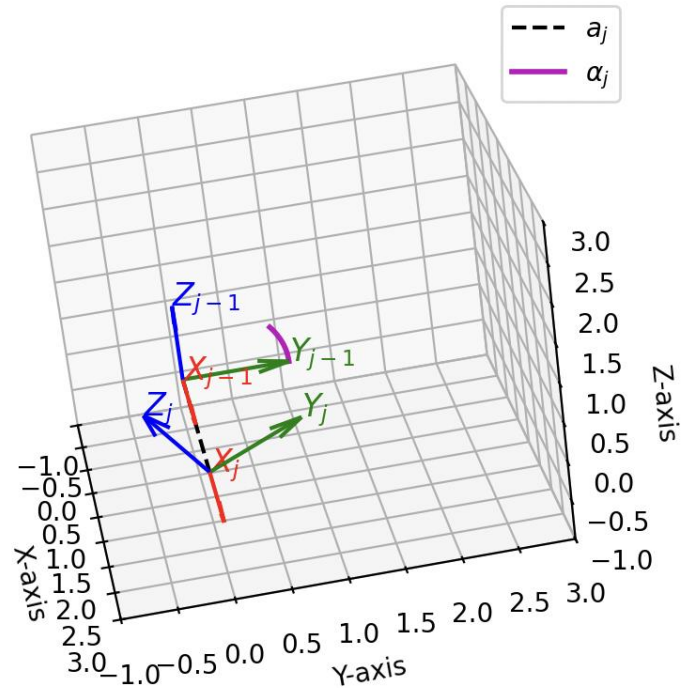


<https://youtu.be/rA9tm0gTln8?si=WQhUdXpkl9JBavcv>

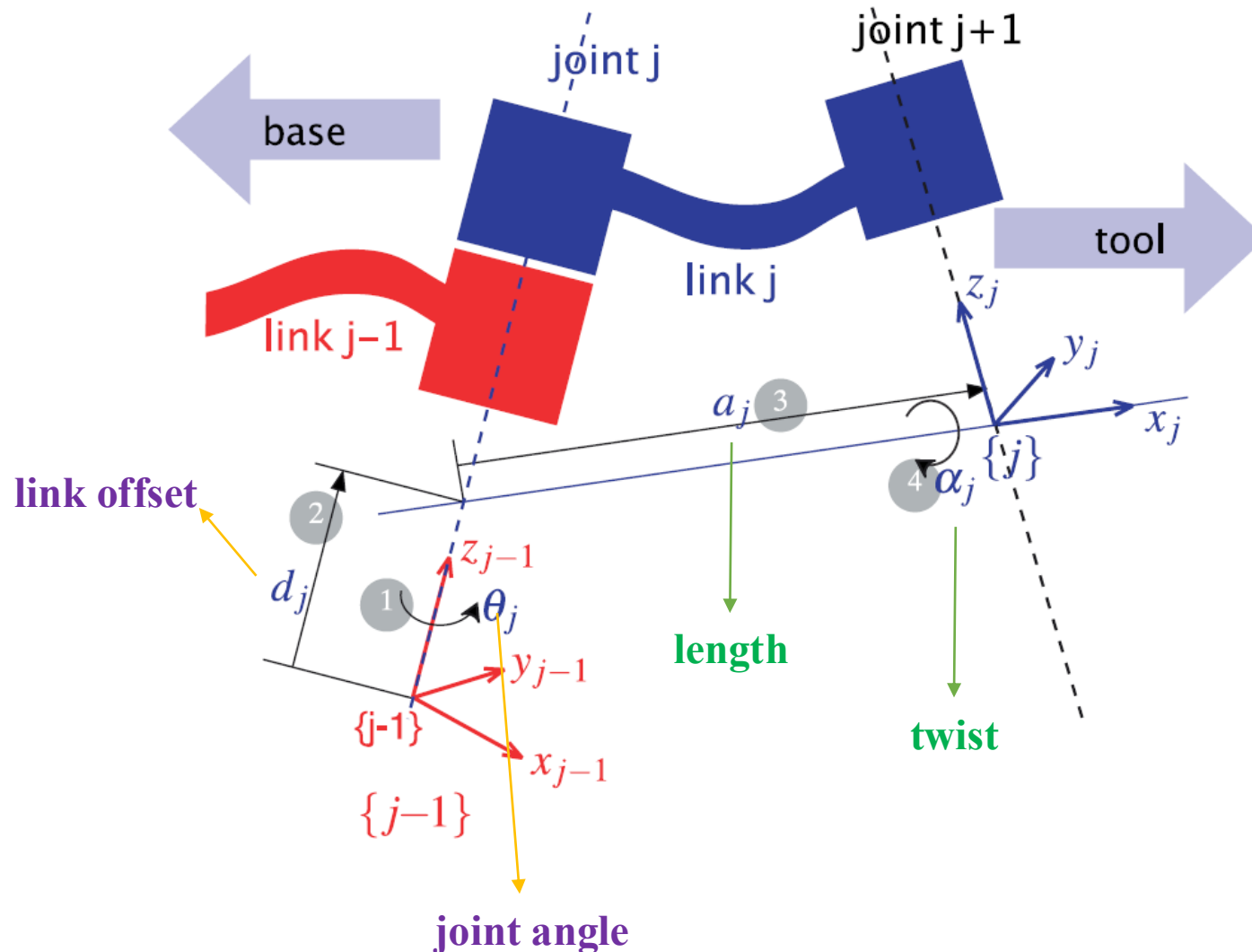


# Twist $\alpha_i$ [see the code `denavit_alpha_a.py`]

Denavit-Hartenberg Convention: Corrected Representation of  $a_j$  and  $\alpha_j$



# Definition of standard Denavit and Hartenberg link parameters



|             |            |                                                                                                                                               |                          |
|-------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Joint angle | $\theta_j$ | the angle between the $x_{j-1}$ and $x_j$ axes about the $z_{j-1}$ axis                                                                       | revolute joint variable  |
| Link offset | $d_j$      | the distance from the origin of frame $j-1$ to the $x_j$ axis along the $z_{j-1}$ axis                                                        | prismatic joint variable |
| Link length | $a_j$      | the distance between the $z_{j-1}$ and $z_j$ axes along the $x_j$ axis; for intersecting axes is parallel to $\hat{z}_{j-1} \times \hat{z}_j$ | constant                 |
| Link twist  | $\alpha_j$ | the angle from the $z_{j-1}$ axis to the $z_j$ axis about the $x_j$ axis                                                                      | constant                 |
| Joint type  | $\sigma_j$ | $\sigma = 0$ for a revolute joint, $\sigma = 1$ for a prismatic joint                                                                         | constant                 |

$$\begin{aligned}
 A_i &= Rot_{z,\theta_i} Trans_{z,d_i} Trans_{x,a_i} Rot_{x,\alpha_i} \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &\quad \times \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

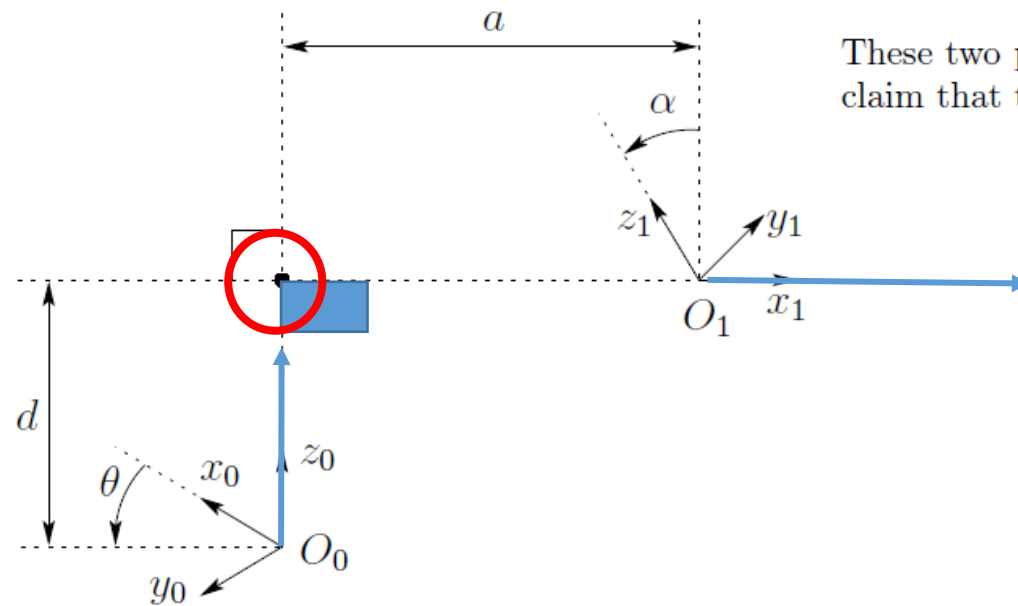


# Definition of standard Denavit and Hartenberg link parameters

## DH Coordinate Frame Assumptions

(DH1) The axis  $x_1$  is perpendicular to the axis  $z_0$ .

(DH2) The axis  $x_1$  intersects the axis  $z_0$ .



These two properties are illustrated in Figure 3.2. Under these conditions, we claim that there exist unique numbers  $a, d, \theta, \alpha$  such that

$$A = Rot_{z,\theta} Trans_{z,d} Trans_{x,a} Rot_{x,\alpha} \quad (3.11)$$

### Three main steps:

1. Define frames
2. Build the DH table
3. Compute matrices

Fig. 3.2 Coordinate frames satisfying assumptions DH1 and DH2

# DH Algorithm

**Step 1.** Locate and label the joint axes  $z_0, \dots, z_{n-1}$

**Step 2.** Establish the base frame. Set the origin anywhere on the  $z_0$  axis. The  $x_0, y_0$  axes are chosen conveniently to form a right-handed frame.

**For**  $i = 1, \dots, n - 1$  perform steps 3 to 5

**Step 3.** Locate the origin  $o_i$  where the common normal to  $z_i$  and  $z_{i-1}$  intersects  $z_i$ .

- If  $z_i$  intersects  $z_{i-1}$  locate  $o_i$  at this intersection.
- If  $z_i$  and  $z_{i-1}$  are parallel, locate  $o_i$  in any convenient position along  $z_i$ .

**Step 4.** If  $z_{i-1}$  and  $z_i$  intersect, establish  $x_i$  in the direction normal to the plane formed by  $z_{i-1}$  and  $z_i$ . In other case, establish  $x_i$  along the common normal between  $z_{i-1}$  and  $z_i$  through  $o_i$ .

**Step 5.** Establish  $y_i$  to complete a right-handed frame.

**Step 6.** Establish the end-effector frame  $o_n x_n y_n z_n$ .

- Assuming the  $n - th$  joint is revolute, set  $z_n = a$  parallel to  $z_{n-1}$ . Establish the origin  $o_n$  conveniently along  $z_n$ , preferably at the center of the gripper or at the tip on any tool that the manipulator may be carrying. Set  $y_n = s$  in the direction of the gripper closure and the set  $x_n = n$  as  $s \times a$ .
- If the tool is not a simple gripper set  $x_n$  and  $y_n$  conveniently to form right-handed frame.

# DH Algorithm

DH Table

**Step 7.** Create a table of DH parameters  $a_i, d_i, \alpha_i, \theta_i$ .

$a_i$  = distance along  $x_i$  from the intersection of the  $x_i$  and the  $z_{i-1}$  axes to  $o_i$ .

$d_i$  = distance along  $z_{i-1}$  from  $o_{i-1}$  to the intersection of the  $x_i$  and  $z_{i-1}$  axes. If joint  $i$  is prismatic,  $d_i$  is variable.

$\alpha_i$  = the angle from  $z_{i-1}$  to  $z_i$  measured about  $x_i$ .

$\theta_i$  = the angle from  $x_{i-1}$  to  $x_i$  measured about  $z_{i-1}$ . If joint  $i$  is revolute,  $\theta_i$  is variable.

Matrices

**Step 8.** From the homogenous transformation matrices  $A_i$  by substituting the above parameters into

$$A_i = Rot_{z,\theta_i} Trans_{z,d_i} Trans_{x,a_i} Rot_{x,\alpha_i}$$

**Step 9.** Form  $T_n^0 = A_1 \cdots A_n$ . This then gives the position and orientation of the tool frame expressed in base coordinates.

# Formula

$$\begin{aligned} A_i &= Rot_{z,\theta_i} Trans_{z,d_i} Trans_{x,a_i} Rot_{x,\alpha_i} \\ &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &\quad \times \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_ic_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_is_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$A_i = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_ic_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_is_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

## Example: Planar Elbow Manipulator

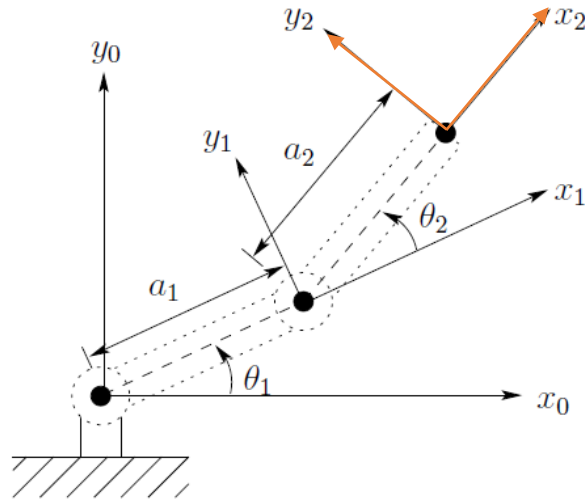


Fig. 3.6 Two-link planar manipulator. The  $z$ -axes all point out of the page, and are not shown in the figure

Table 3.1 Link parameters for 2-link planar manipulator

| Link | $a_i$ | $\alpha_i$ | $d_i$ | $\theta_i$   |
|------|-------|------------|-------|--------------|
| 1    | $a_1$ | 0          | 0     | $\theta_1^*$ |
| 2    | $a_2$ | 0          | 0     | $\theta_2^*$ |

\* variable

**Step 1.** Locate and label the joint axes  $z_0, \dots, z_{n-1}$

**Step 2.** Establish the base frame. Set the origin anywhere on the  $z_0$  axis. The  $x_0, y_0$  axes are chosen conveniently to form a right-handed frame.

**For  $i = 1, \dots, n - 1$  perform steps 3 to 5**

**Step 3.** Locate the origin  $o_i$  where the common normal to  $z_i$  and  $z_{i-1}$  intersects  $z_i$ .

- If  $z_i$  intersects  $z_{i-1}$  locate  $o_i$  at this intersection.
- If  $z_i$  and  $z_{i-1}$  are parallel, locate  $o_i$  in any convenient position along  $z_i$ .

**Step 4.** Establish  $x_i$  along the common normal between  $z_{i-1}$  and  $z_i$  through  $o_i$ , or in the direction normal to the  $z_{i-1} - z_i$  plane if  $z_{i-1}$  and  $z_i$  intersect.

**Step 5.** Establish  $y_i$  to complete a right-handed frame.

**Step 6.** Establish the end-effector frame  $o_n x_n y_n z_n$ .

- Assuming the  $n - th$  joint is revolute, set  $z_n = a$  parallel to  $z_{n-1}$ . Establish the origin  $o_n$  conveniently along  $z_n$ , preferably at the center of the gripper or at the tip on any tool that the manipulator may be carrying. Set  $y_n = s$  in the direction of the gripper closure and the set  $x_n = n$  as  $s \times a$ .
- If the tool is not a simple gripper set  $x_n$  and  $y_n$  conveniently to form right-handed frame.

**Step 7.** Create a table of DH parameters  $a_i, d_i, \alpha_i, \theta_i$ .

$a_i$  = distance along  $x_i$  from the intersection of the  $x_i$  and the  $z_{i-1}$  axes to  $o_i$ .

$d_i$  = distance along  $z_{i-1}$  from  $o_{i-1}$  to the intersection of the  $x_i$  and  $z_{i-1}$  axes. If joint  $i$  is prismatic,  $d_i$  is variable.

$\alpha_i$  = the angle from  $z_{i-1}$  to  $z_i$  measured about  $x_i$ .

$\theta_i$  = the angle from  $x_{i-1}$  to  $x_i$  measured about  $z_{i-1}$ . If joint  $i$  is revolute,  $\theta_i$  is variable.

**Step 8.** Compute the homogenous transformation matrices  $A_i$  by substituting the above parameters into

$$A_i = Rot_{z, \theta_i} Trans_{z, d_i} Trans_{x, a_i} Rot_{x, \alpha_i}$$

**Step 9.** Compute  $T_n^0 = A_1 \cdots A_n$ . This then gives the position and orientation of the tool frame expressed in base coordinates

## Example: Planar Elbow Manipulator

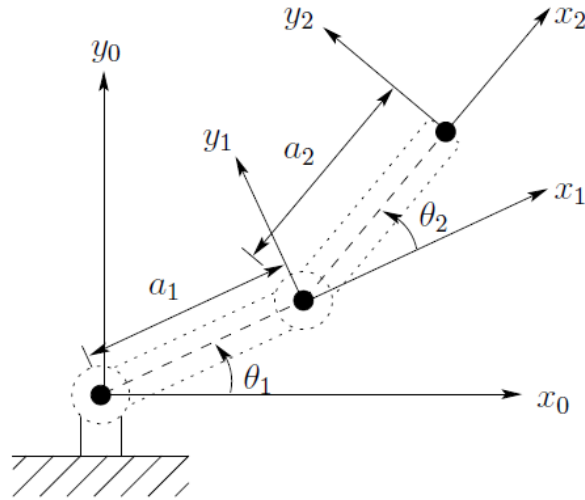


Fig. 3.6 Two-link planar manipulator. The  $z$ -axes all point out of the page, and are not shown in the figure

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|------|-------|------------|-------|--------------|
| 1    | $a_1$ | 0          | 0     | $\theta_1^*$ |
| 2    | $a_2$ | 0          | 0     | $\theta_2^*$ |

\* variable

Formula:  $A_i = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The  $T$ -matrices are thus given by

$$T_1^0 = A_1$$

$$T_2^0 = A_1 A_2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Notice that the first two entries of the last column of  $T_2^0$  are the  $x$  and  $y$  components of the origin  $o_2$  in the base frame; that is,

$$x = a_1 c_1 + a_2 c_{12}$$

$$y = a_1 s_1 + a_2 s_{12}$$

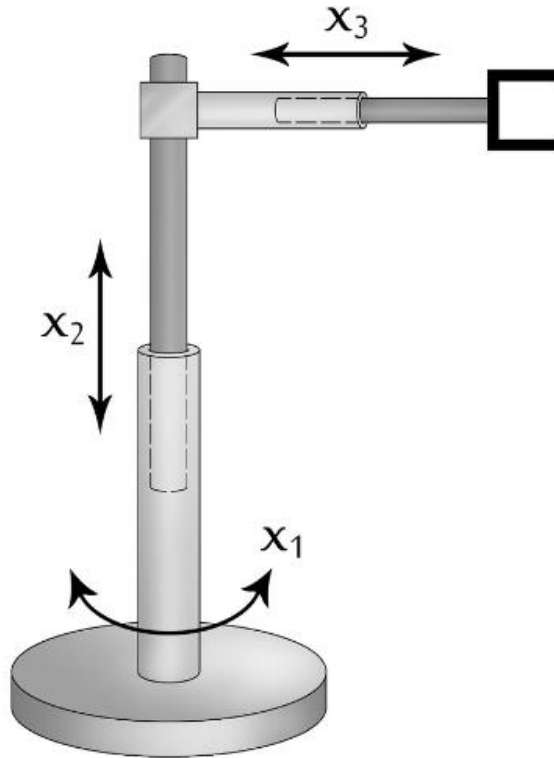
are the coordinates of the end-effector in the base frame. The rotational part of  $T_2^0$  gives the orientation of the frame  $o_2 x_2 y_2 z_2$  relative to the base frame.

◇

# Three-Link Cylindrical Robot

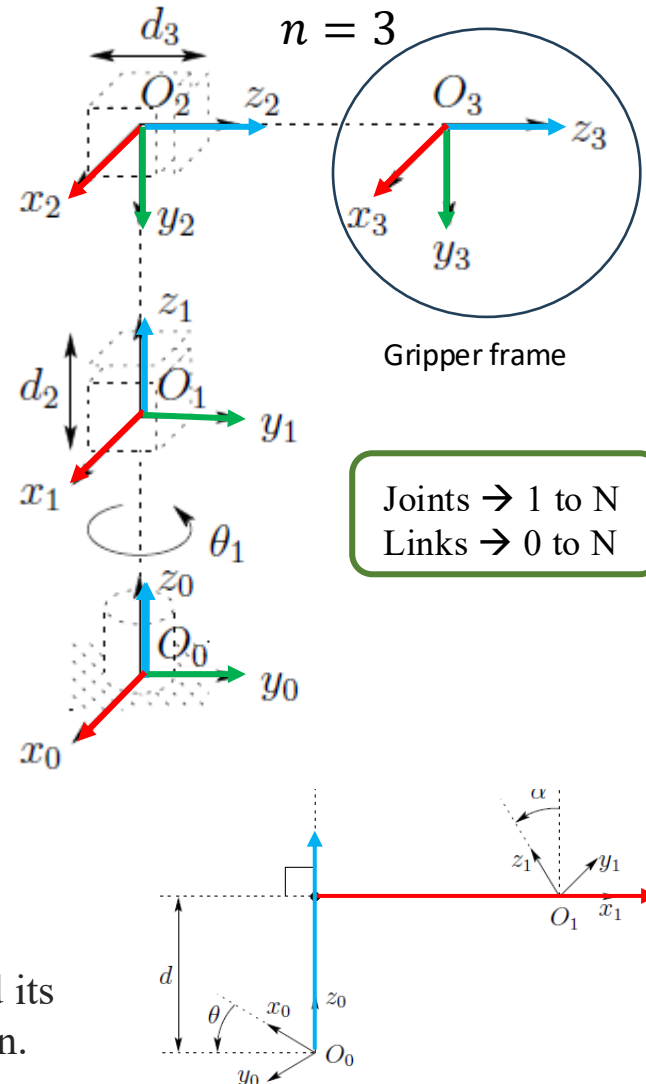
| Link | $a_i$ | $\alpha_i$ | $d_i$   | $\theta_i$   |
|------|-------|------------|---------|--------------|
| 1    | 0     | 0          | $d_1^*$ | $\theta_1^*$ |
| 2    | 0     | $-90$      | $d_2^*$ | 0            |
| 3    | 0     | 0          | $d_3^*$ | 0            |

\* variable



## Remember:

- Joint  $i$  is fixed w.r.t. frame  $i$ .
- When joint  $i$  is actuated, link  $i$  and its attached frame experience a motion.



## Simplified steps:

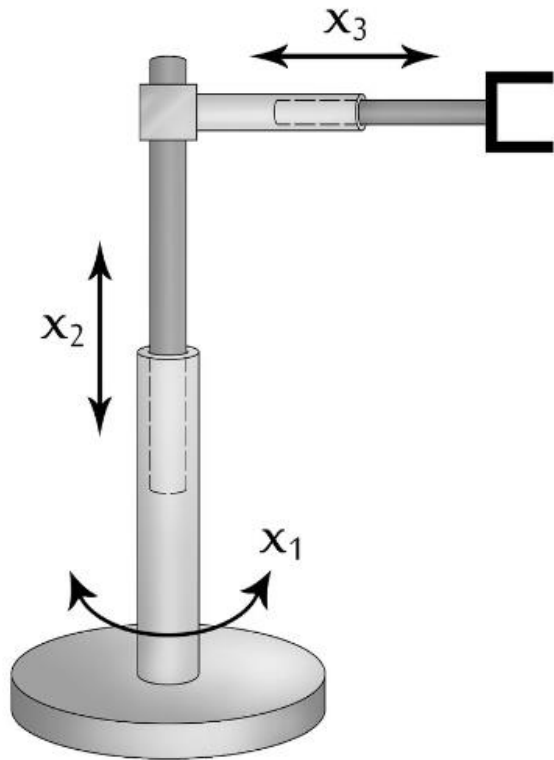
- Define  $n$  according to the degrees of freedom of robot.
- Assign the  $z$  axes such that  $z_i$  to be the axis of actuation for joint  $i + 1$ .
- Define the base frame in a convenient manner.
- Choose the  $x_i$  with Step 4 of the algorithm:
  - If  $z_{i-1}$  and  $z_i$  intersect, establish  $x_i$  in the direction normal to the plane formed by  $z_{i-1}$  and  $z_i$ . In other case, establish  $x_i$  along the common normal between  $z_{i-1}$  and  $z_i$  through  $o_i$ .
- Corroborate the 2 assumptions for all frames.

### DH Coordinate Frame Assumptions

- (DH1) The axis  $x_1$  is perpendicular to the axis  $z_0$ .  
 (DH2) The axis  $x_1$  intersects the axis  $z_0$ .

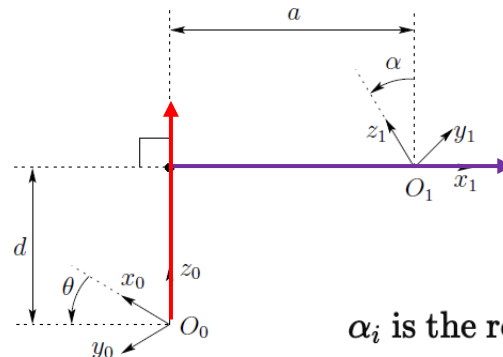
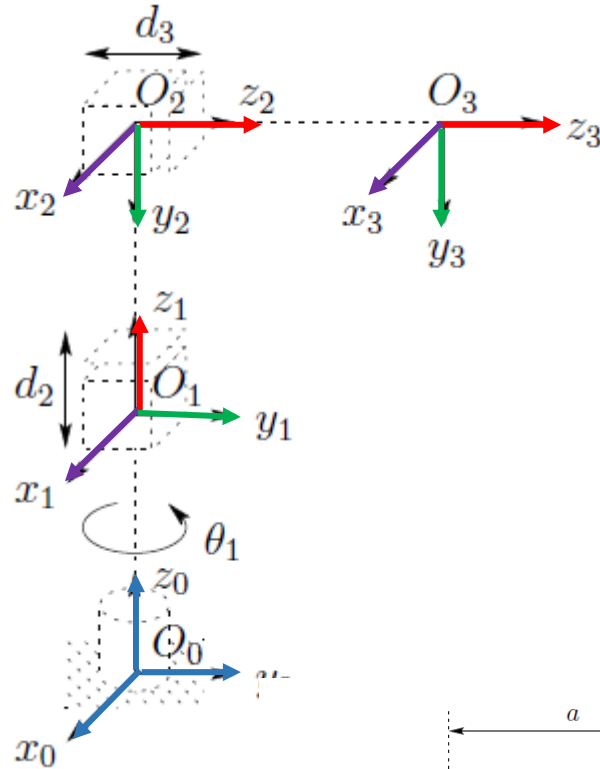
- Establish  $y_i$  to complete a right-handed frame

# Three-Link Cylindrical Robot



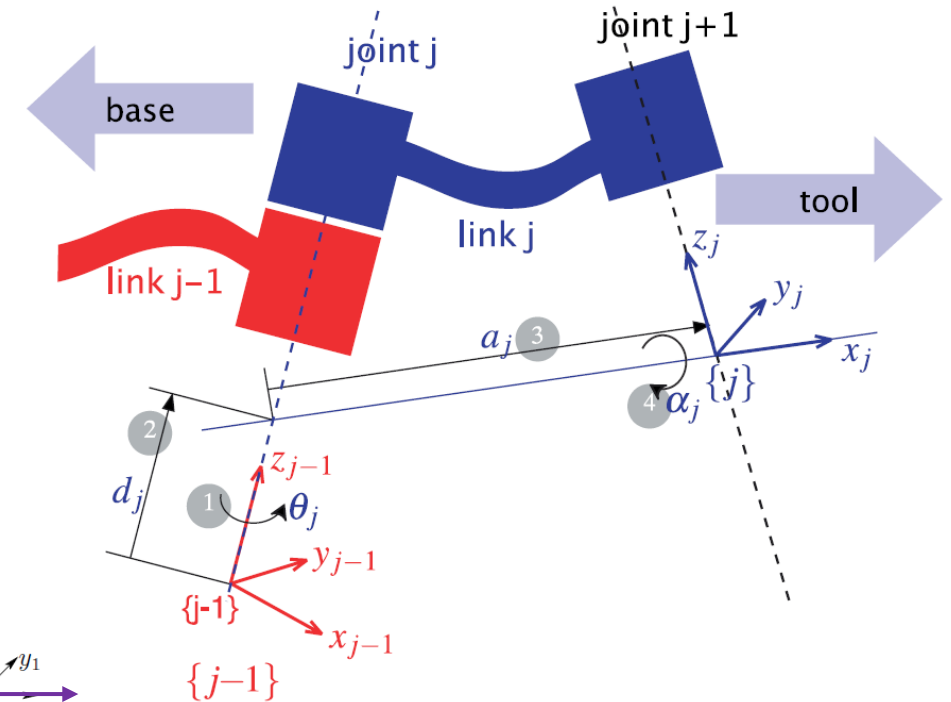
| Link | $a_i$ | $\alpha_i$ | $d_i$   | $\theta_i$   |
|------|-------|------------|---------|--------------|
| 1    | 0     | 0          | $d_1$   | $\theta_1^*$ |
| 2    | 0     | $-90$      | $d_2^*$ | 0            |
| 3    | 0     | 0          | $d_3^*$ | 0            |

\* variable



## Simplified steps:

- We create the table of DH parameters. Rows of the table = n [degrees of freedom]



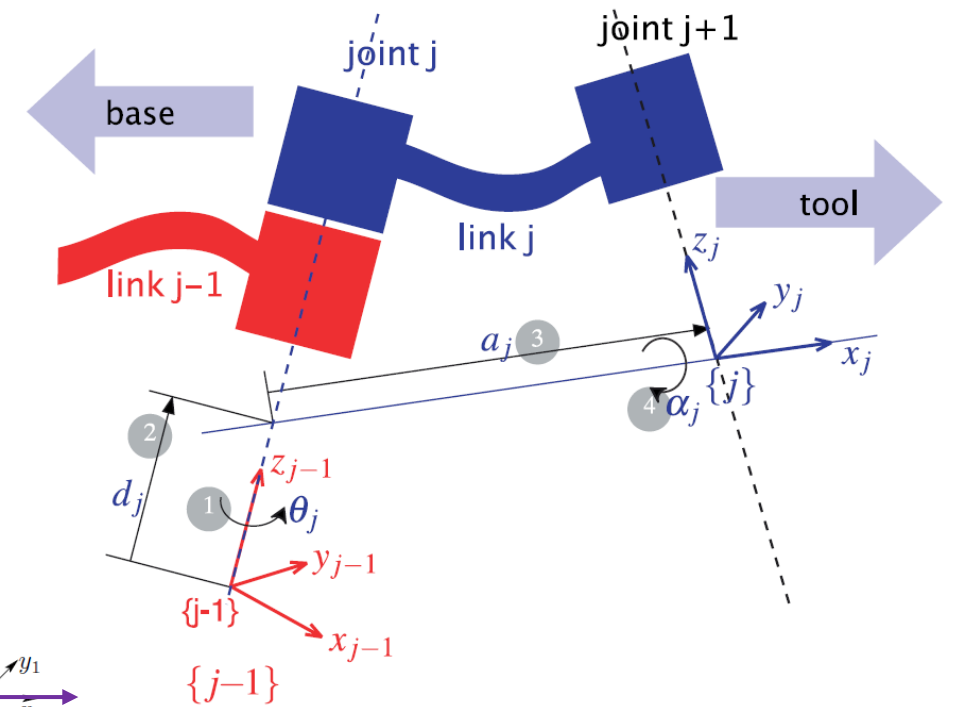
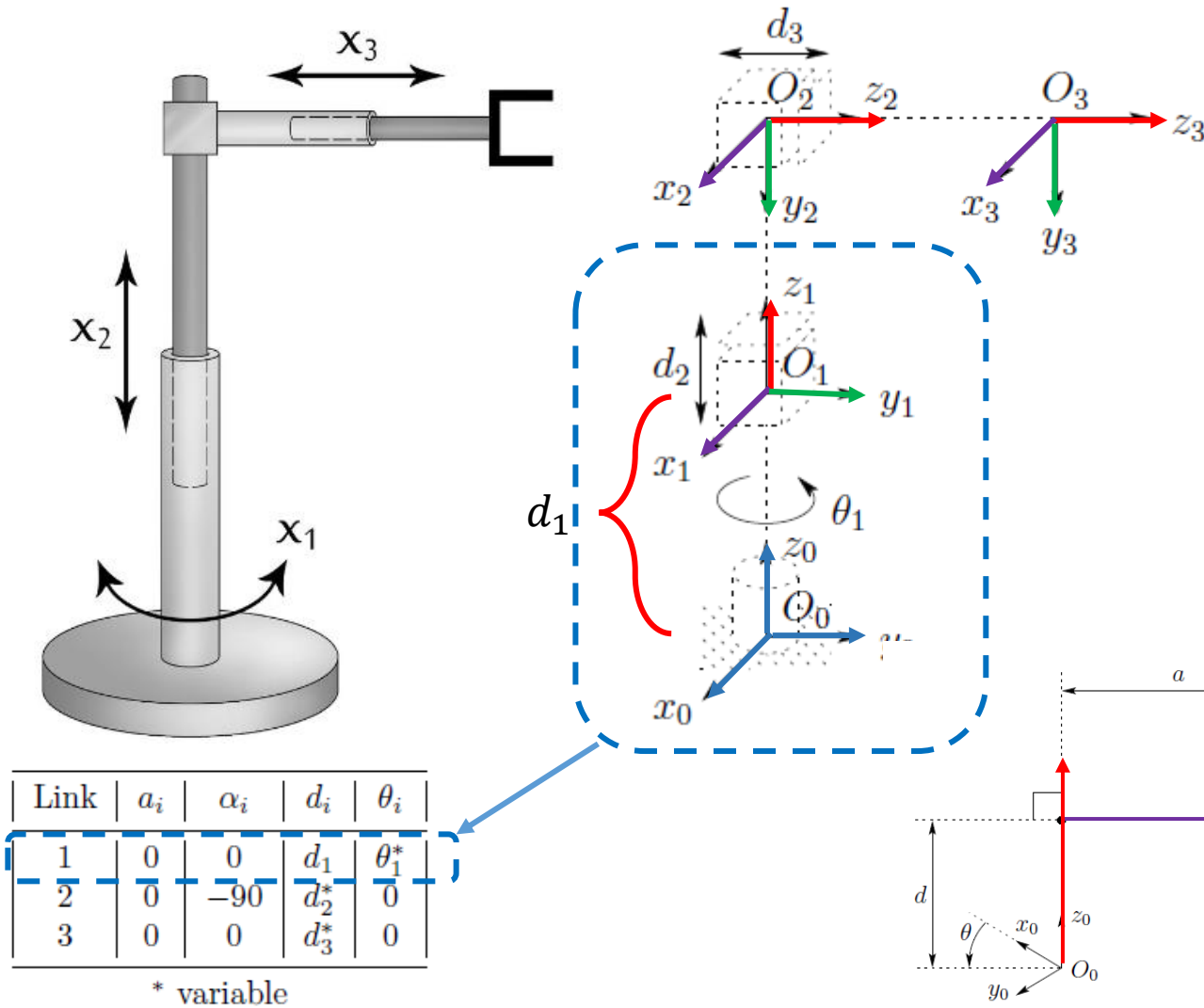
$\alpha_i$  is the rotation angle about the  $x_i$  axis that aligns  $z_{i-1}$  with  $z_i$ .



# Three-Link Cylindrical Robot

**Simplified steps:**

8. We create the table of DH parameters.

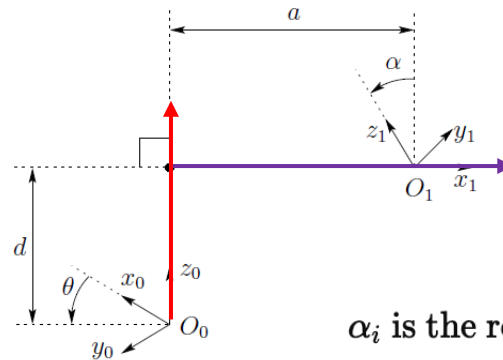
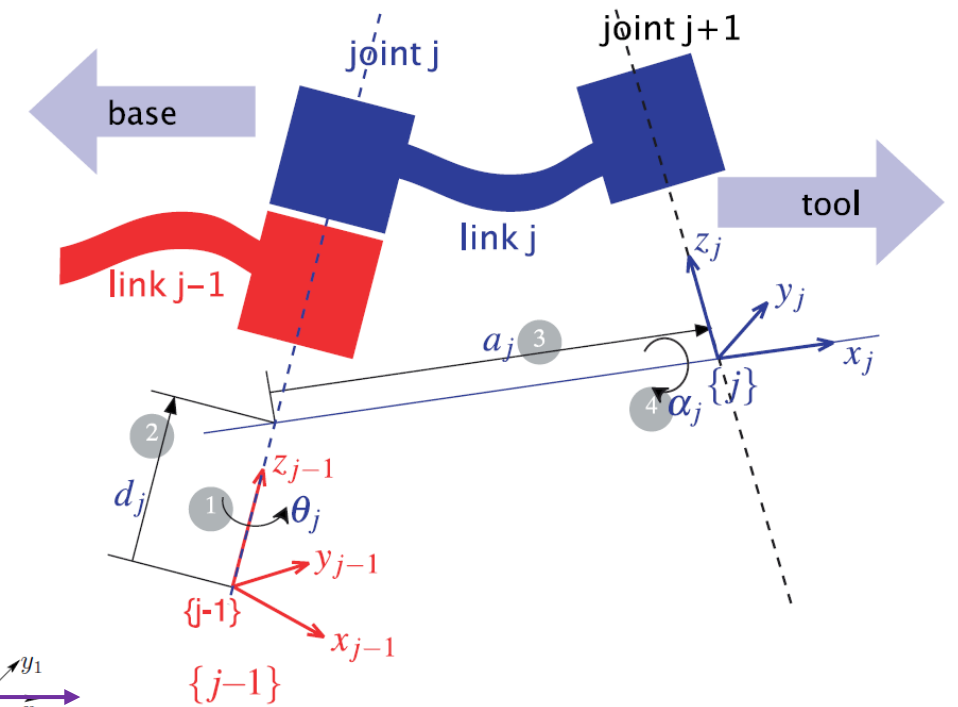
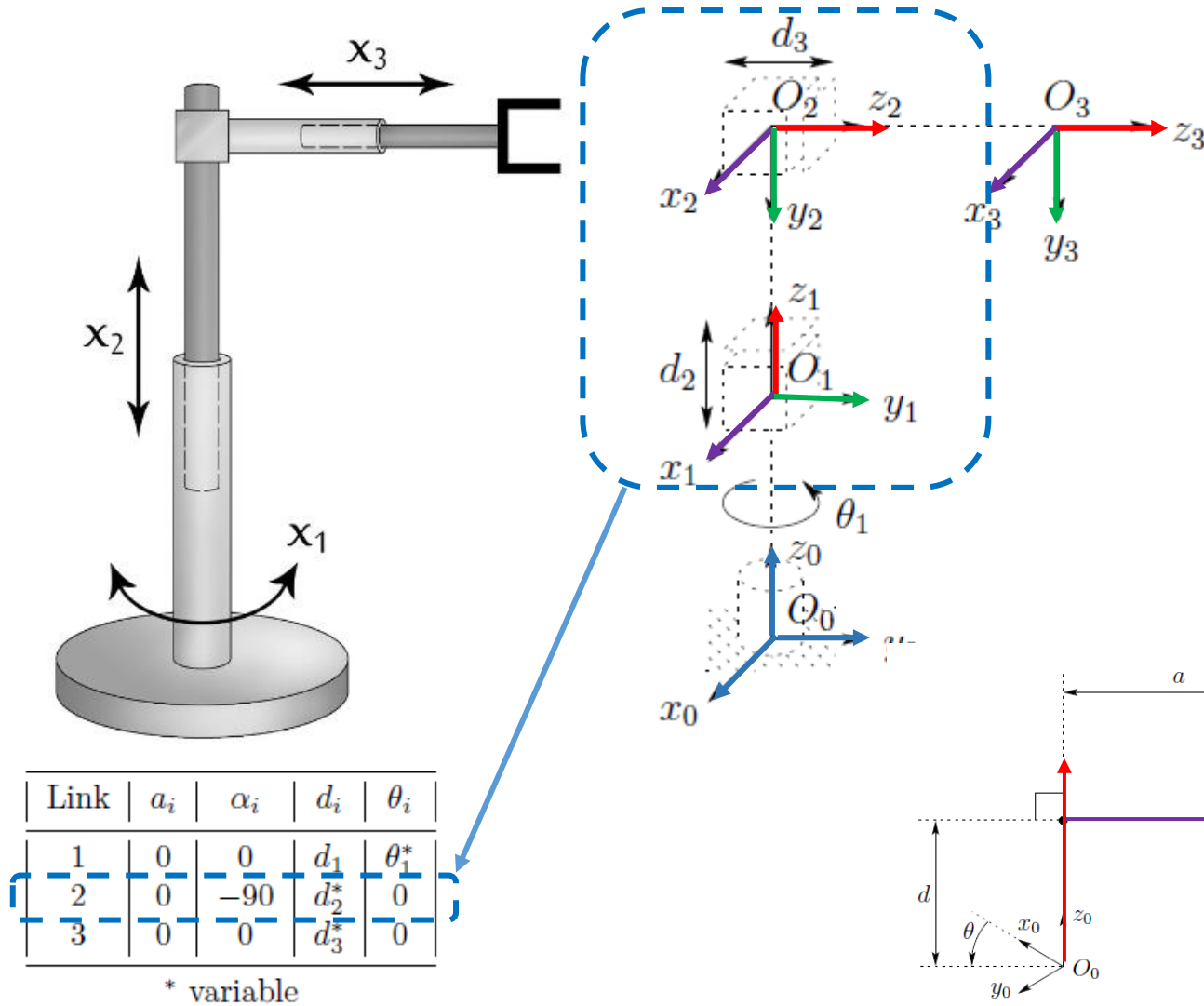


$\alpha_i$  is the rotation angle about the  $x_i$  axis that aligns  $z_{i-1}$  with  $z_i$ .

# Three-Link Cylindrical Robot

**Simplified steps:**

8. We create the table of DH parameters.

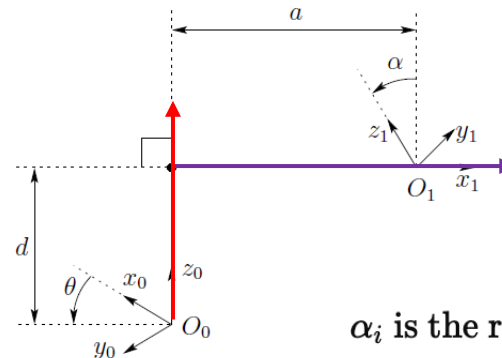
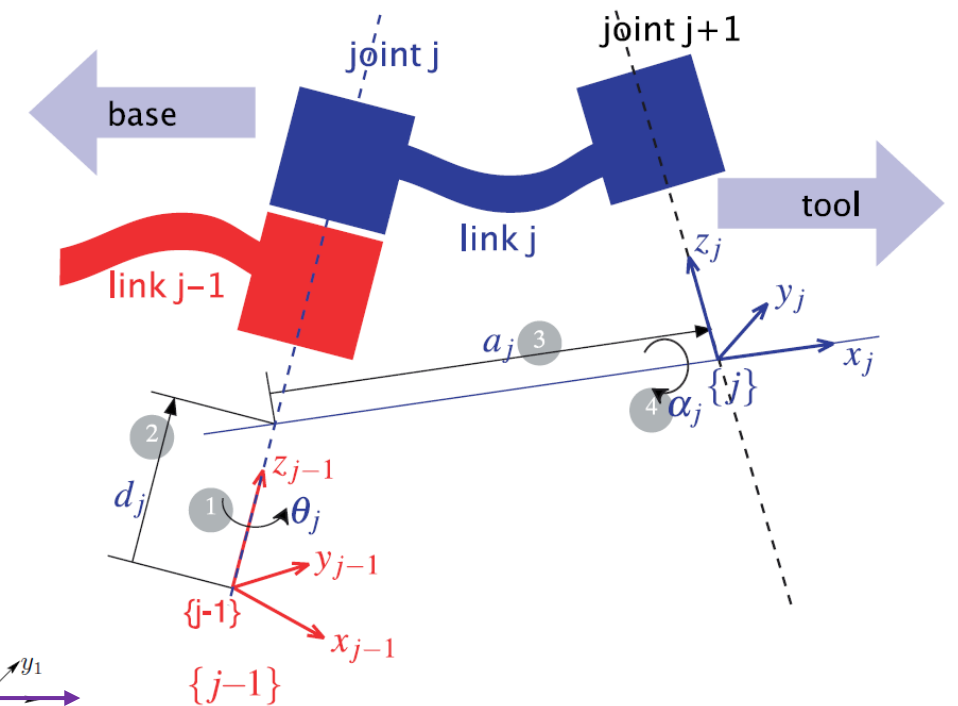
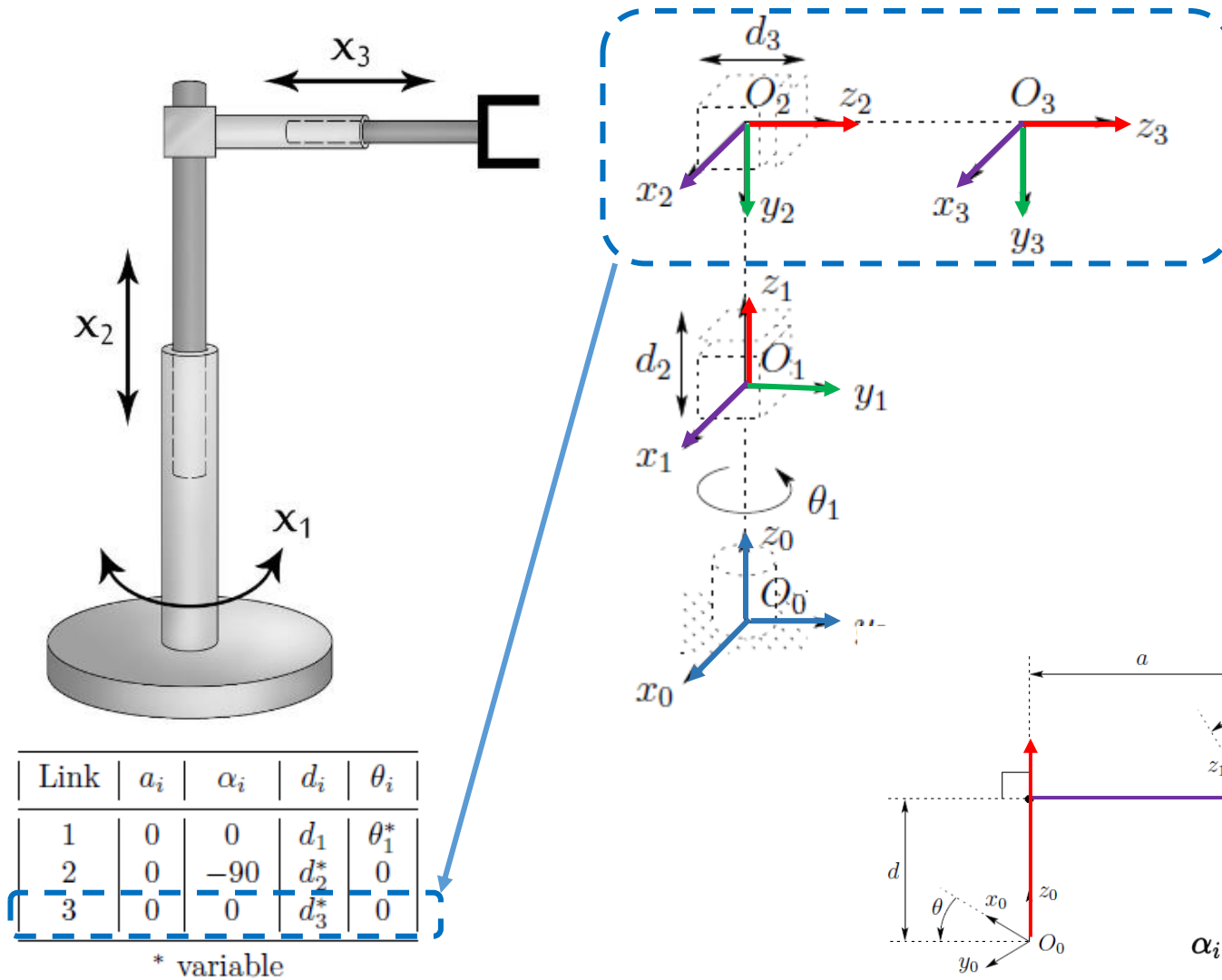


$\alpha_i$  is the rotation angle about the  $x_i$  axis that aligns  $z_{i-1}$  with  $z_i$ .

# Three-Link Cylindrical Robot

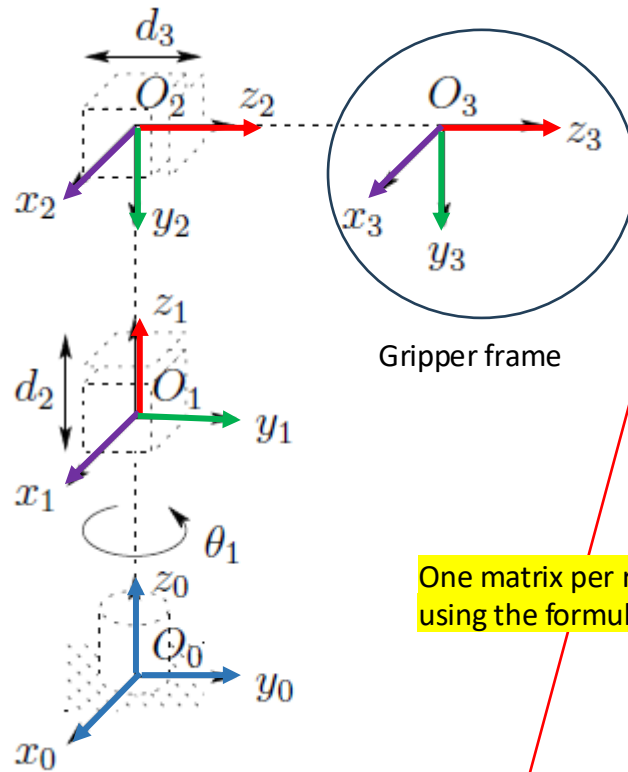
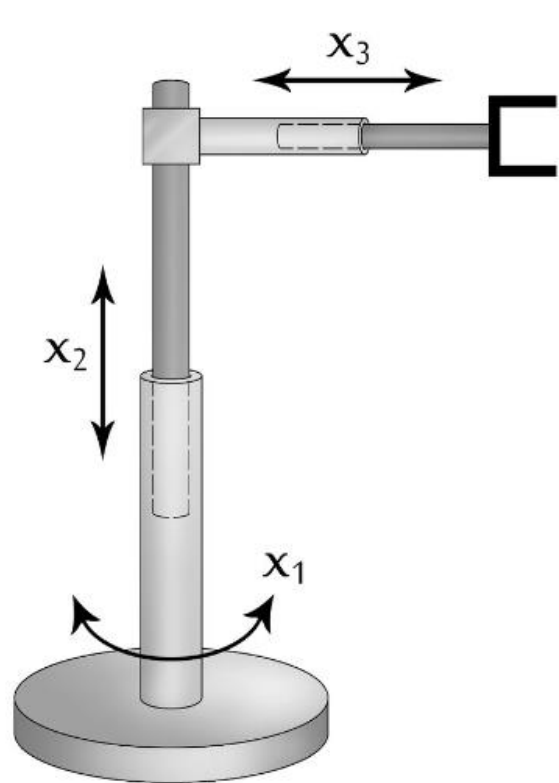
**Simplified steps:**

8. We create the table of DH parameters.



$\alpha_i$  is the rotation angle about the  $x_i$  axis that aligns  $z_{i-1}$  with  $z_i$ .

**Formula:**  $A_i = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_ic_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_is_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$



Gripper frame

One matrix per row  
using the formula

| Link | $a_i$ | $\alpha_i$ | $d_i$   | $\theta_i$   |
|------|-------|------------|---------|--------------|
| 1    | 0     | 0          | $d_1$   | $\theta_1^*$ |
| 2    | 0     | $-90$      | $d_2^*$ | 0            |
| 3    | 0     | 0          | $d_3^*$ | 0            |

\* variable

**Simplified steps:**  
9. Build the matrices

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

10. Build the  $T_n^0$

$$T_3^0 = A_1 A_2 A_3 = \begin{bmatrix} c_1 & 0 & -s_1 & -s_1 d_3 \\ s_1 & 0 & c_1 & c_1 d_3 \\ 0 & -1 & 0 & d_1 + d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Spherical wrist

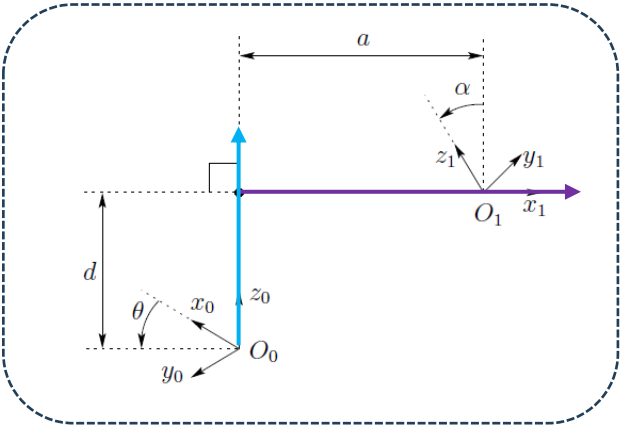
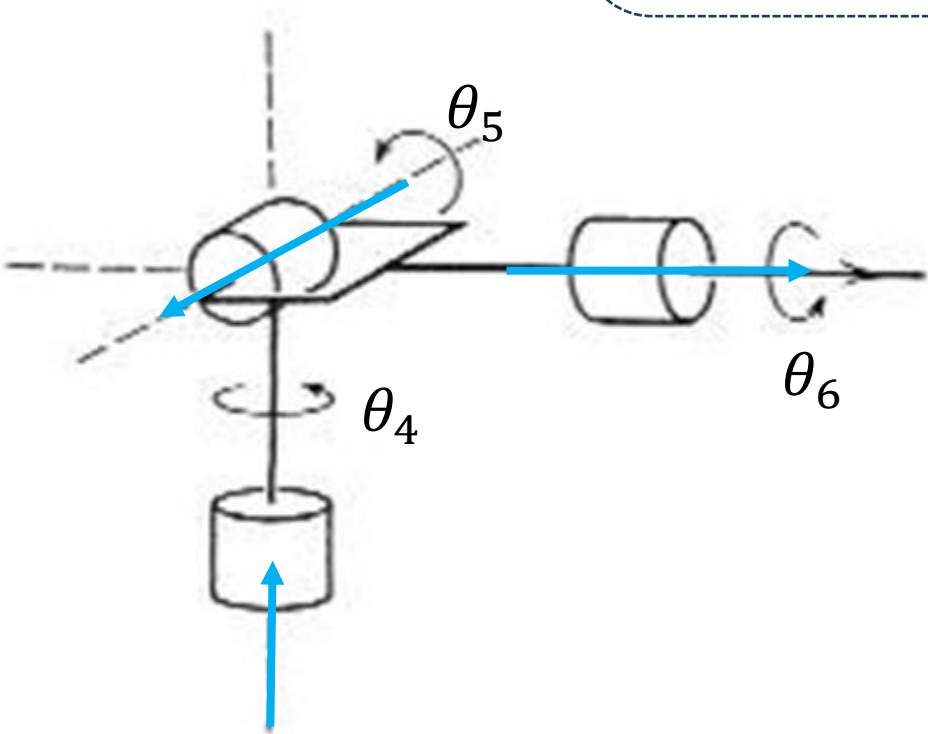


Table 3.3 DH parameters for spherical wrist

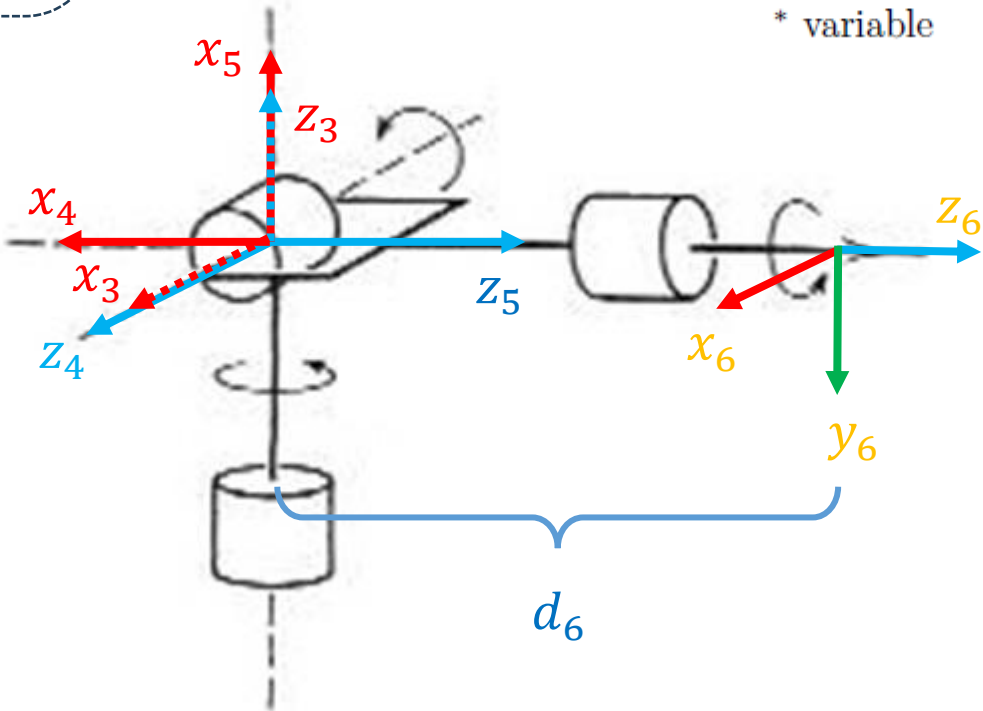
| Link | $a_i$ | $\alpha_i$ | $d_i$ | $\theta_i$   |
|------|-------|------------|-------|--------------|
| 4    | 0     | -90        | 0     | $\theta_4^*$ |
| 5    | 0     | 90         | 0     | $\theta_5^*$ |
| 6    | 0     | 0          | $d_6$ | $\theta_6^*$ |

Use the right-hand rule where your thumb is  $x_4$

\* variable



Notice the direction of each arrow.



We have intentionally omitted the y-axis for clarity. Please try to identify the relevant values.

$\alpha_i$  is the rotation angle about the  $x_i$  axis that aligns  $z_{i-1}$  with  $z_i$ .

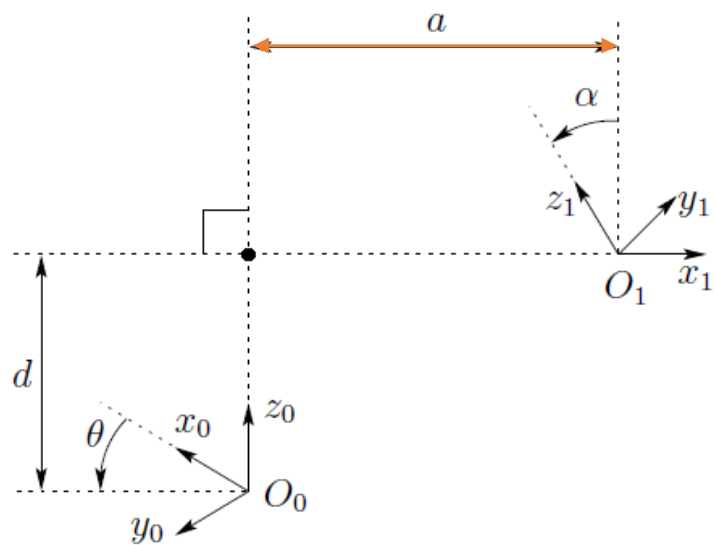
# Spherical wrist

$$\begin{aligned}
 A_4 &= \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 A_5 &= \begin{bmatrix} c_5 & 0 & s_5 & 0 \\ s_5 & 0 & -c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 A_6 &= \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

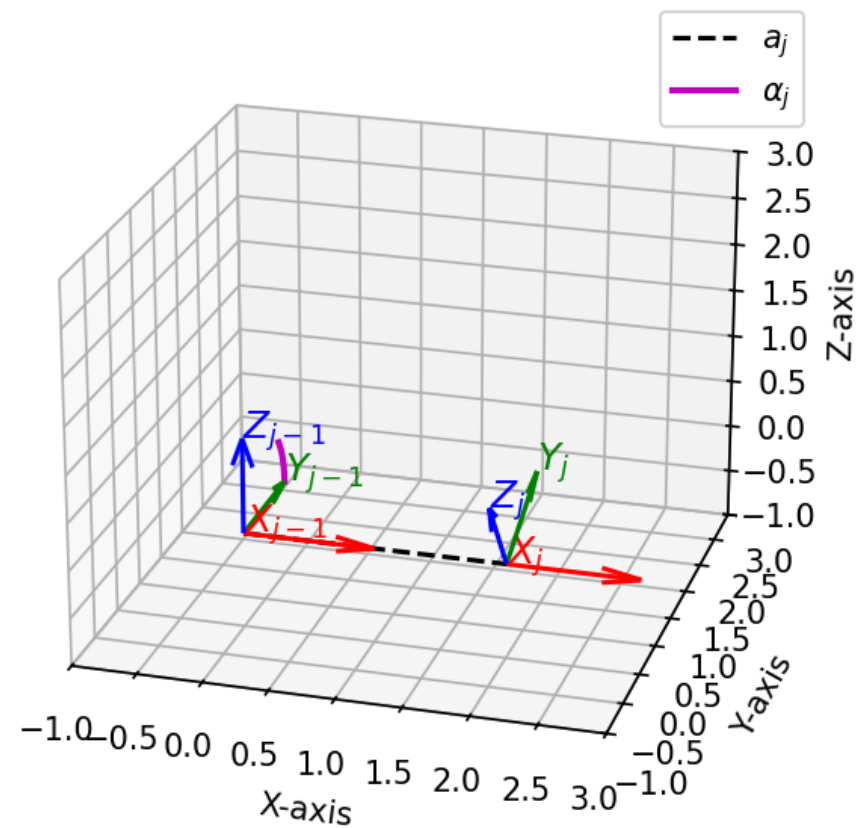
*Multiplying these together yields*

$$\begin{aligned}
 T_6^3 &= A_4 A_5 A_6 \\
 &= \begin{bmatrix} R_6^3 & o_6^3 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

# Recap



Denavit-Hartenberg Convention: Corrected Representation of  $a_j$  and  $\alpha_j$



See code!

# Recap

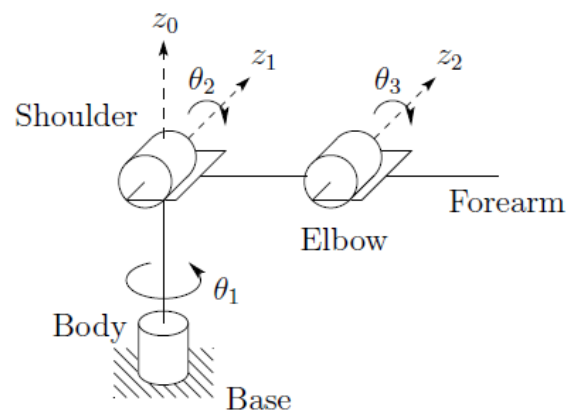


Fig. 1.8 Structure of the elbow manipulator.

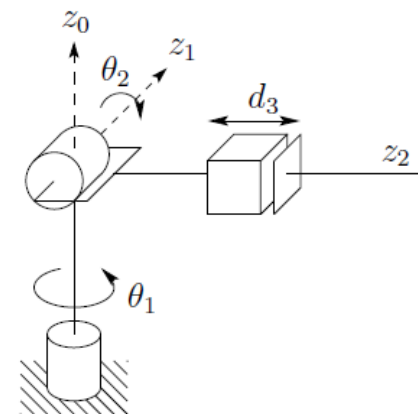
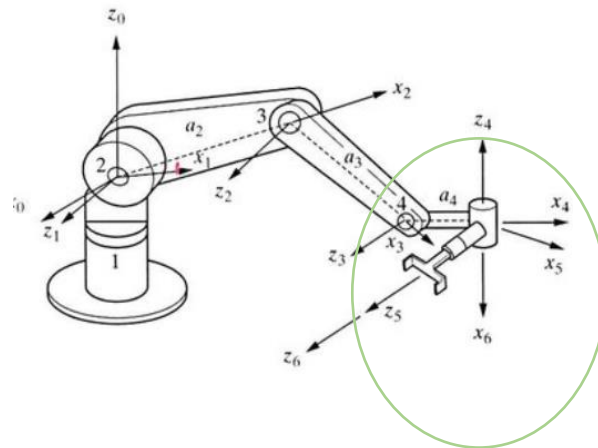


Fig. 1.10 The spherical manipulator.

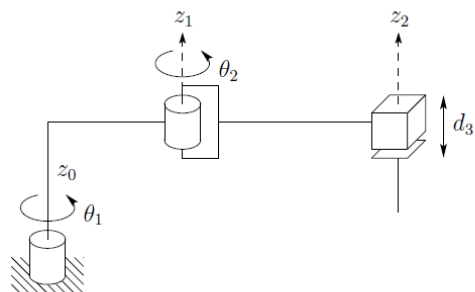


Fig. 1.13 The SCARA (Selective Compliant Articulated Robot for Assembly).

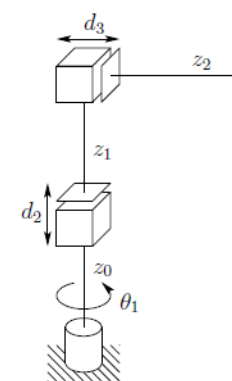


Fig. 1.16 The cylindrical manipulator.





# Visual Studio Code

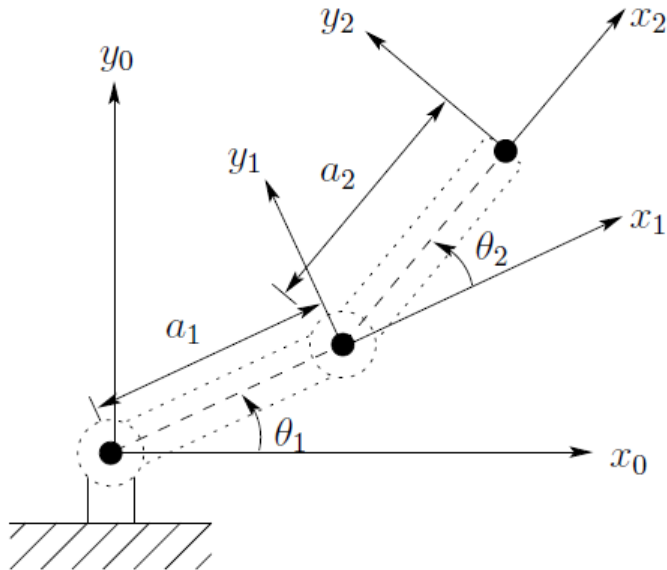
Install Visual Studio Code

Follow the steps:

- <https://code.visualstudio.com/docs/python/python-tutorial>
- Install Peter Corke's Robotics library:
- <https://github.com/petercorke/robotics-toolbox-python>

# Exercise 1

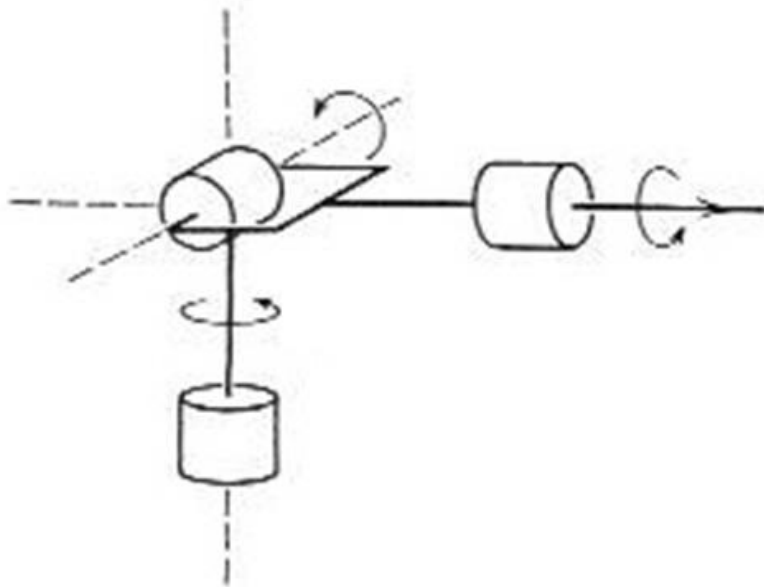
- Problem:
  1. Following the previous analysis compute the  $T_n^0$  for the planar robot, and give the coordinates of the tip.
  2. Verify your results on Python using the Corke's libraries. Study the codes:
    1. `robotarmDH.py` and `robotarmAnalytics-3DCylindrical.py`



*Fig. 3.6* Two-link planar manipulator. The  $z$ -axes all point out of the page, and are not shown in the figure

# Exercise 2

- Problem:
  1. Following the previous analysis compute the  $T_n^0$  for the robot below, and give the coordinates of the tip.
  2. Verify your results on Python using the Corke's libraries.



# Exercise 3

- Problem:

1. Following the previous analysis compute the  $T_n^0$  for the robot below, and give the coordinates of the tip.
2. Verify your results on Python using the Corke's libraries.

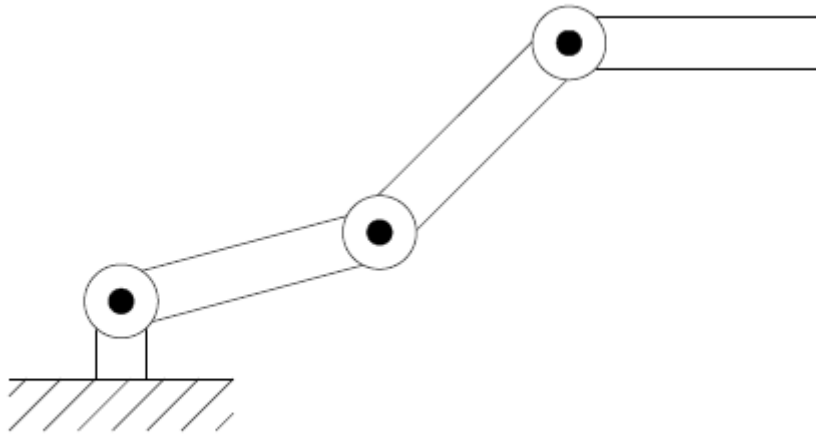


Fig. 3.23 Three-link planar arm of Problem 3.2

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# Exercise 4

- Problem:
  1. Following the previous analysis compute the  $T_n^0$  for the robot below, and give the coordinates of the tip.
  2. Verify your results on Python using the Corke's libraries.

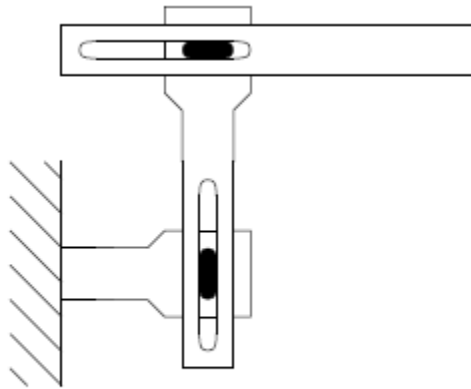
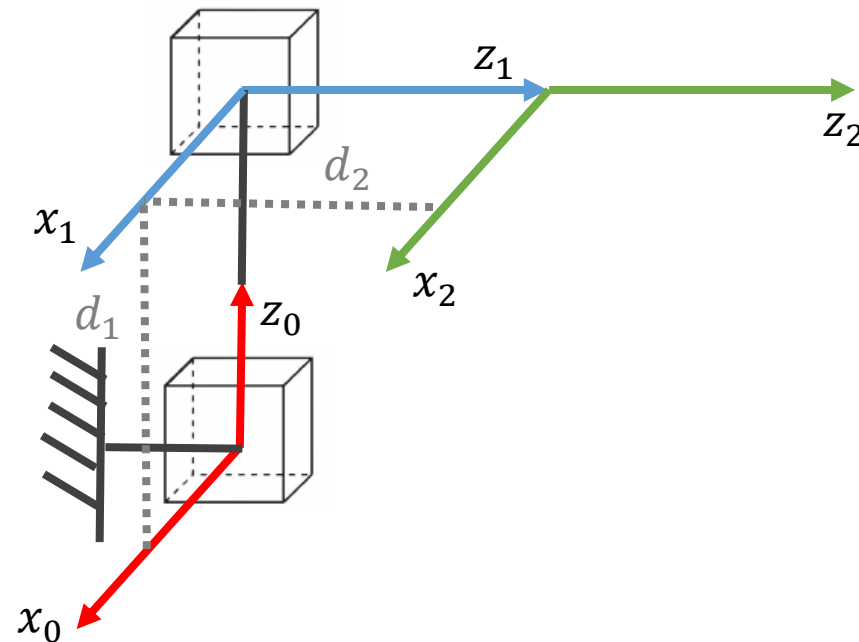


Fig. 3.24 Two-link cartesian robot of Problem 3.3

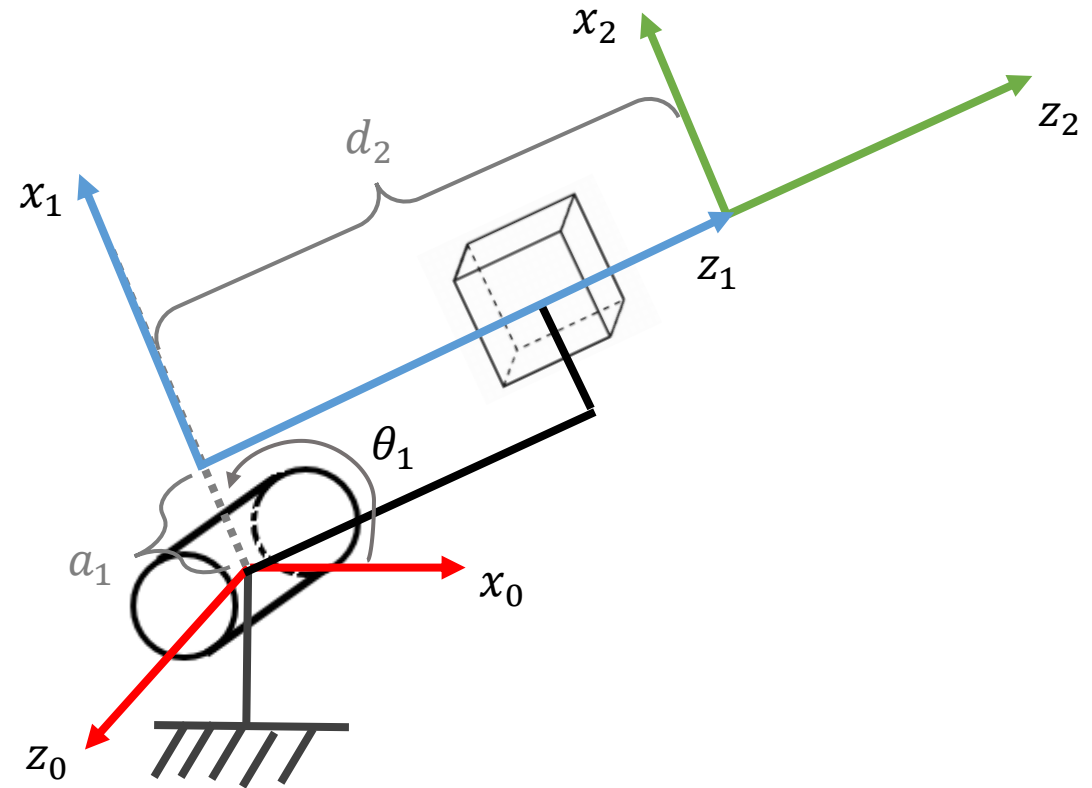
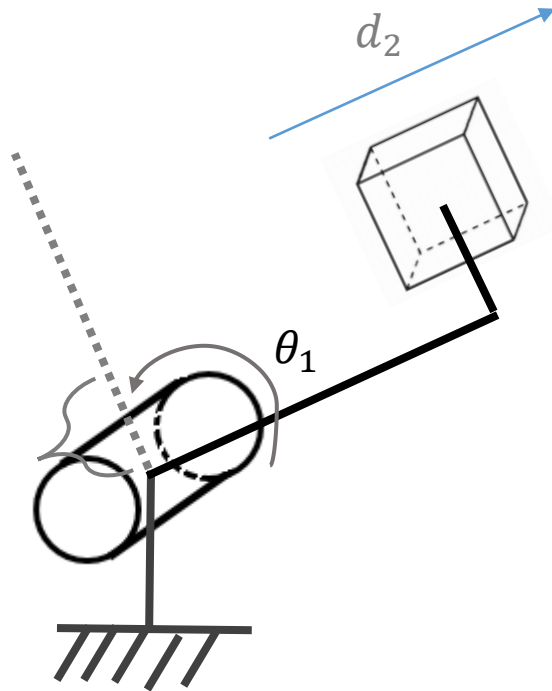


# Exercise 5

| link | $a_i$ | $\alpha_i$ | $d_i$   | $\theta_i$   |
|------|-------|------------|---------|--------------|
| 0    | $a_1$ | 90         | 0       | $\theta_1 *$ |
| 1    | 0     | 0          | $d_2 *$ | 0            |

- Problem:

1. Following the previous analysis compute the  $T_n^0$  for the robot below, and give the coordinates of the tip.



# Forward kinematics (recap)

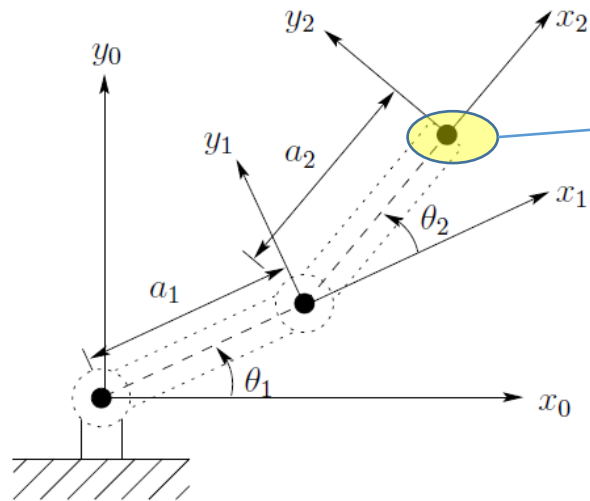


Fig. 3.6 Two-link planar manipulator. The  $z$ -axes all point out of the page, and are not shown in the figure

Orientation ( $R$ )

Position ( $p$ )

$$T_2^0 = A_1 A_2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Notice that the first two entries of the last column of  $T_2^0$  are the  $x$  and  $y$  components of the origin  $o_2$  in the base frame; that is,

$$\begin{aligned} x &= a_1 c_1 + a_2 c_{12} \\ y &= a_1 s_1 + a_2 s_{12} \end{aligned}$$

are the coordinates of the end-effector in the base frame. The rotational part of  $T_2^0$  gives the orientation of the frame  $o_2 x_2 y_2 z_2$  relative to the base frame.

◇

With the forward kinematics process we find the position (**vector  $p$** ) and orientation (**matrix  $R$** ) of the tip of the robot.

# Inverse Kinematics

The general problem of inverse kinematics can be stated as follows. Given a  $4 \times 4$  homogeneous transformation

$$H = \begin{bmatrix} R & o \\ 0 & 1 \end{bmatrix} \in SE(3) \quad (3.31)$$

with  $R \in SO(3)$ , find (one or all) solutions of the equation

$$T_n^0(q_1, \dots, q_n) = H \quad (3.32)$$

where

$$T_n^0(q_1, \dots, q_n) = A_1(q_1) \cdots A_n(q_n) \quad (3.33)$$

Here,  $H$  represents the desired position and orientation of the end-effector, and our task is to find the values for the joint variables  $q_1, \dots, q_n$  so that  $T_n^0(q_1, \dots, q_n) = H$ .

So,  $R_d$  and  $p_d$  are given, then we have:

$$\begin{pmatrix} R & p \\ 0, 0, 0 & 1 \end{pmatrix} = \begin{pmatrix} R_d & p_d \\ 0, 0, 0 & 1 \end{pmatrix}.$$

And we need to find the values of each element of  $(R, p)$ .



# Inverse Kinematics Example



Stanford Manipulator

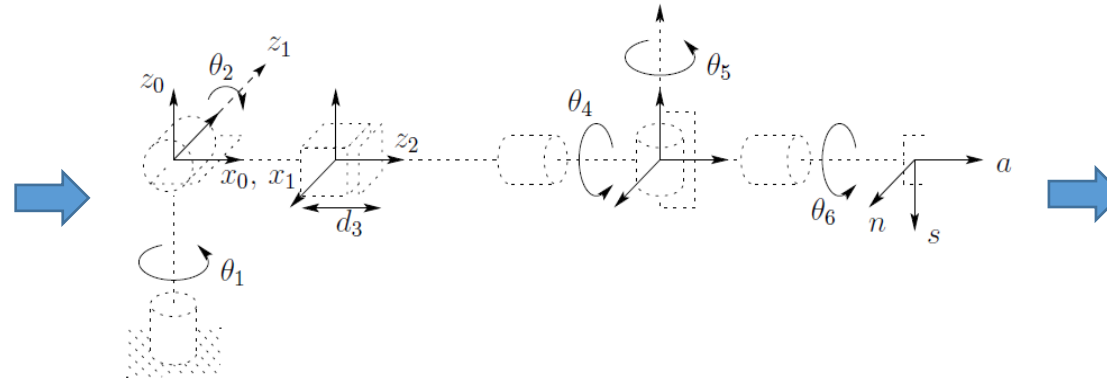


Table 3.4 DH parameters for Stanford Manipulator

| Link | $d_i$ | $a_i$ | $\alpha_i$ | $\theta_i$ |
|------|-------|-------|------------|------------|
| 1    | 0     | 0     | -90        | $\theta^*$ |
| 2    | $d_2$ | 0     | +90        | $\theta^*$ |
| 3    | $d^*$ | 0     | 0          | 0          |
| 4    | 0     | 0     | -90        | $\theta^*$ |
| 5    | 0     | 0     | +90        | $\theta^*$ |
| 6    | $d_6$ | 0     | 0          | $\theta^*$ |

\* joint variable

Fig. 3.10 DH coordinate frame assignment for the Stanford manipulator

$$A_1 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_5 = \begin{bmatrix} c_5 & 0 & s_5 & 0 \\ s_5 & 0 & -c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$T_6^0$  is then given as

$$T_6^0 = A_1 \cdots A_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & d_x \\ r_{21} & r_{22} & r_{23} & d_y \\ r_{31} & r_{32} & r_{33} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left\{ \begin{array}{l} r_{11} = c_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] - d_2(s_4c_5c_6 + c_4s_6) \\ r_{21} = s_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] + c_1(s_4c_5c_6 + c_4s_6) \\ r_{31} = -s_2(c_4c_5c_6 - s_4s_6) - c_2s_5c_6 \\ r_{12} = c_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] - s_1(-s_4c_5s_6 + c_4c_6) \\ r_{22} = -s_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] + c_1(-s_4c_5s_6 + c_4c_6) \\ r_{32} = s_2(c_4c_5s_6 + s_4c_6) + c_2s_5s_6 \\ r_{13} = c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5 \\ r_{23} = s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5 \\ r_{33} = -s_2c_4s_5 + c_2c_5 \\ d_x = c_1s_2d_3 - s_1d_2 + d_6(c_1c_2c_4s_5 + c_1c_5s_2 - s_1s_4s_5) \\ d_y = s_1s_2d_3 + c_1d_2 + d_6(c_1s_4s_5 + c_2c_4s_1s_5 + c_5s_1s_2) \\ d_z = c_2d_3 + d_6(c_2c_5 - c_4s_2s_5) \end{array} \right.$$

# Inverse Kinematics Example



Stanford Manipulator

Suppose that the desired position and orientation of the final frame are given by:

$$H = \begin{bmatrix} 0 & 1 & 0 & -0.154 \\ 0 & 0 & 1 & 0.763 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To find the corresponding joint variables  $\theta_1, \theta_2, d_3, \theta_4, \theta_5$  and  $\theta_6$  we must solve the following simultaneous set of nonlinear trigonometric equations:

$$c_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] - s_1(s_4c_5c_6 + c_4s_6) = 0$$

$$s_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] + c_1(s_4c_5c_6 + c_4s_6) = 0$$

$$-s_2(c_4c_5c_6 - s_4s_6) - c_2s_5c_6 = 1$$

$$c_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] - s_1(-s_4c_5s_6 + c_4c_6) = 1$$

$$s_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] + c_1(-s_4c_5s_6 + c_4c_6) = 0$$

$$s_2(c_4c_5s_6 + s_4c_6) + c_2s_5s_6 = 0$$

$$c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5 = 0$$

$$s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5 = 1$$

$$-s_2c_4s_5 + c_2c_5 = 0$$

$$c_1s_2d_3 - s_1d_2 + d_6(c_1c_2c_4s_5 + c_1c_5s_2 - s_1s_4s_5) = -0.154$$

$$s_1s_2d_3 + c_1d_2 + d_6(c_1s_4s_5 + c_2c_4s_1s_5 + c_5s_1s_2) = 0.763$$

$$c_2d_3 + d_6(c_2c_5 - c_4s_2s_5) = 0$$

# Inverse Kinematics Example

*If the values of the nonzero DH parameters are  $d_2 = 0.154$  and  $d_6 = 0.263$ , one solution to this set of equations is given by:*

$$\theta_1 = \pi/2, \quad \theta_2 = \pi/2, \quad d_3 = 0.5, \quad \theta_4 = \pi/2, \quad \theta_5 = 0, \quad \theta_6 = \pi/2.$$

*Even though we have not yet seen how one might derive this solution, it is not difficult to verify that it satisfies the forward kinematics equations for the Stanford arm.*

## **Problem:**

Verify the solution on Python.

# How to solve the Inverse Kinematics problem?

- Numerically! Using the **Levenberg–Marquardt** algorithm.
- Simple code:

```
import roboticstoolbox as rtb
from spatialmath import SE3
import matplotlib.pyplot as plt

# Define the robot's DH parameters (theta, d, a, alpha)
# Example of a robot with 3 links
L1 = rtb.RevoluteDH(d=0.5, a=1, alpha=0) # Link 1
L2 = rtb.RevoluteDH(d=0, a=1, alpha=0) # Link 2
L3 = rtb.RevoluteDH(d=0, a=1, alpha=0) # Link 3

# Create the robot from the links
robot = rtb.DHRobot([L1, L2, L3], name='3-DOF Robot')

# Display the DH table (this prints the DH parameters)
print(robot)

# Joint configuration (angles in radians for each link)
q = [0, 0, 0] # You can modify the values to change the robot's posture

# Calculate forward kinematics (the final transformation of the end-effector)
T = robot.fkine(q)
print(f"End-effector position (forward kinematics): \n{T}")
```

```
# Inverse kinematics
```

```
# Define the desired pose for the end-effector
```

```
Tep = SE3.Trans(0, 0, 3.5) # Target pose (x=1, y=0.8, z=0.5)
```

```
# Solve inverse kinematics to find the joint angles for the desired pose
```

```
q_inv = robot.ikine_LM(Tep) # Levenberg-Marquardt IK solver
```

```
# Extract the joint configuration that achieves the desired pose
```

```
q_solution = q_inv.q
```

```
print(f"Joint angles (inverse kinematics solution): {q_solution}")
```

```
# Plot the robot in the configuration obtained from inverse kinematics
```

```
robot.plot(q_solution, block=True)
```

```
# Interactive tool to visualize the robot's joint states and DH parameters
```

```
robot.teach(jointlabels=1) # Opens an interactive window with DH axes
```